Signal Table Per State

Power On

Signal	I/O	State
reset_n	Input	LOW
CKE	Input	LOW
CK_t	Input	Toggling

Reset

Signal	I/O	State
reset_n	Input	HIGH — releases reset
CKE	Input	HIGH — enables clock

Bank Idle

Signal	I/O	State
CKE	Input	HIGH
ACT	Input	HIGH — triggers transition to Activating

Activating

Signal	I/O	State
CA	Input	Carries bank + row address
cmd_valid_int	Internal	1-cycle pulse — triggers Bank Active transition
tRCD_cnt	Internal	Counting down — must expire before Bank Active
open_row_id	Internal	Saves row address for later validation

Bank Active

Signal	I/O	State
CA	Input	Receives RD/WR/PRE command
row_buffer	Internal	Holds open row data
bank_state	Internal	FSM ready state
tRCD_flag	Internal	HIGH — confirms tRCD expired
open_row_id	Internal	Validates incoming command targets correct row

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Read

Signal	I/O	State
CA	Input	READ command + column address
tCL_cnt	Internal	Counting down CAS latency
dq_out_reg	Internal	Loaded with data to output
RD_DATA_VALID	Output	1-cycle pulse — HIGH when data ready
DQ	Output	Drives read data to PHY
DQS_t/c	Output	Differential strobe — toggling during data burst

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Write

Signal	I/O	State
CA	Input	WRITE command + column address
DQ	Input	Incoming write data from PHY
DQS_t/c	Input	PHY driven strobe — DUT samples DQ on edges
DM	Input	Byte mask — LOW=write, HIGH=skip
wl_pipe_cnt	Internal	Write latency countdown
dq_capture	Internal	DQ latched on DQS edge

Signal	I/O	State
row_buffer	Internal	Updated with masked write data

Precharging

Signal	I/O	State
CA	Input	PRECHARGE command
tRP_cnt	Internal	Counting down — must expire before IDLE
row_buffer	Internal	Written back to memory array
ROW_OPEN	Internal	Cleared to LOW — row marked closed
bank_state	Internal	Transitions to IDLE after tRP expires